

Autonomous Robotics: Final Project

SLAM, Exploration, WiFi Mapping and Autodocking with TurtleBot

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Core Mission Workflow:

- Localization and Mapping
- Frontier-based Exploration
- Path Planning & Following
- WiFi Signal Mapping
- Autonomous Return to Base & Docking

Localization & Mapping

- **LiDAR:** Scans the forward environment to detect obstacles and room geometry as the robot progresses.
- `slam_toolbox` **Package:** Create an occupancy grid and locate the robot in the created map



Frontier Selection Heuristic

Frontier Point: a point separating known free space from unknown space.

Optimal Point: selected as the one minimizing the following heuristic:

Heuristic Equation

$$score(f) = w_1 \cdot D_{robot}(f) - w_2 \cdot N_{unknown}(f)$$

- $N_{unknown}(f)$: Number of unknown cells to be discovered around f .
- $D_{robot}(f)$: Euclidean distance from the robot to f .

Continuous exploration: Frontier points are recomputed every 10s. If the optimal point is unreachable, it is removed from the list, and a new one is computed.

Path Planning & Following

1. Planning

- Obstacles are expanded to the robot's radius.
- Calculates the optimal path using **A***.
- Replanning after a fixed amount of time.

2. Following

- A virtual "carrot" moves along the path.
- It slows down if it isn't close enough to the robot.
- Replanning if the carrot has been waiting for too long.



Carrot Waiting for Robot

Start the Exploration Mission

The main mission executes as a sequential, autonomous pipeline:

1. **Undocking:** The robot detaches from the charging station and rotates to the open space.
2. **Exploration Trigger:** A service is called to officially start the autonomous exploration node.
3. **Execution:** The robot utilizes the Frontier Heuristic and A* to explore the space until a stop condition is met.

Exploration Stop Conditions

The autonomous exploration loop evaluates mission status continuously and terminates under three primary conditions:

1. **Low Battery Level:** Hardware threshold triggered. The robot aborts exploration to ensure enough power remains for docking.
2. **Map Completion:** No reachable frontier points left (nothing left to explore in the bounded environment).
3. **Time Out:** Mission duration exceeds the maximum allowed operational time limit.

Autonomous Docking & Base Return

- **Return to Dock:** When a stop condition is reached, the robot uses *PlanToEssential* to go to a saved position near the docking station, then uses *GoToPose* to reach it more precisely.
- **Autodock Mechanism:**
 - Uses the default Kobuki autodocking system.
 - Laser sensors help align the robot with the dock.
 - The robot docks automatically.
 - If charging does not start, it backs up and tries again.

WiFi Signal Mapping

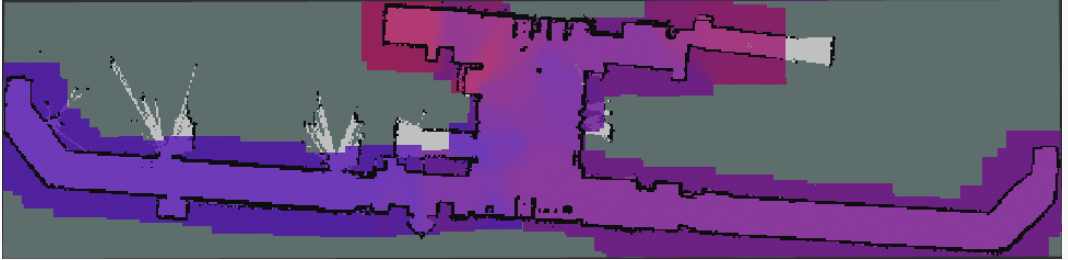
- **WiFi Scanning:** All WiFi networks are scanned using `wpa_cli`, and only one selected BSSID is used to build the final map.
- Each measurement contributes to nearby cells using a weighted window:

$$S(x, y) = \frac{\sum_i w_i(x, y) s_i}{\sum_i w_i(x, y)}$$

- s_i : RSSI value measured at robot position
- $w_i(x, y)$: distance-based Gaussian weight

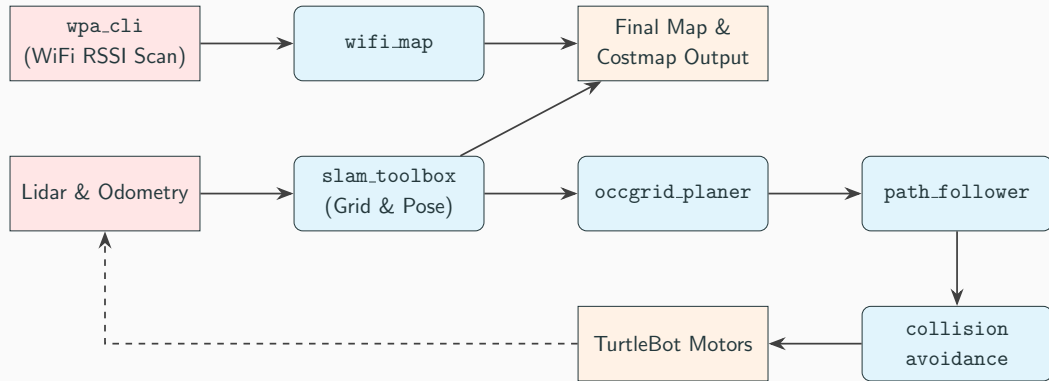
⇒ Nearby measurements have stronger influence on the map

Results: Integrated WiFi Map



Final generated occupancy map overlaid with targeted WiFi Signal Strength.

System Architecture & Workflow



Questions?